

Channel-Aware Cooperative Perception for Blind Spot Detection in V2X Networks

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Abstract

Autonomous vehicles have become a focal point of research and development because of their ability to perceive their surroundings. However, physical obstructions and limited perceptions create blind spots onboard sensors cannot resolve alone. This report proposes a channel-aware cooperative vehicle-infrastructure perception system that addresses this issue by using a Roadside Unit (RSU) to detect objects in blind spots and selectively transmit the information to nearby vehicles using a bandwidth-constrained Vehicle-to-Everything (V2X) channel.

The system operates as an end-to-end pipeline on the DAIR-V2X dataset, combining YOLOv10-N object detection, ByteTrack multi-object tracking, and bird's eye view localization using monocular depth and LiDAR calibration. A Hungarian matching algorithm with a depth-aware Mahalanobis distance cost matrix identifies blind spot targets, and a lightweight Multi-Layer Perceptron (MLP) Semantic Gate prioritizes which targets to transmit under realistic per-frame bandwidth budgets.

Comparative testing demonstrates the MLP Semantic Gate consistently outperforms Hard Gate, No Gate, and Random baselines across all transmission levels, achieving a precision-recall score of 0.551 compared to a random baseline of 0.023. The results highlight that prioritizing bandwidth meaningfully improves cooperative perception under real-world communication constraints.

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1 Introduction

Cars driving through busy intersections can only see what their sensors can reach. Objects such as buildings, trucks, other cars consistently block the sensors view, creating blind spots that can lead to accidents. This issue can not just be fixed by adding more sensors to the car because something will always remain hidden.

A practical answer to this problem is to integrate RSUs, cameras and sensors mounted at intersections because they are able to scan the environment transmit the information to nearby vehicles, preventing collisions.

The challenge of cooperative perception is not scanning the surroundings, but the communication between the RSUs and the vehicles. An RSU might detect up to fourty objects in a busy intersection per frame but the V2X channel has limited bandwich, meaning that the RSU cannot send everything. What the RSU decides to transmit can not be random and must consider what the vehicle already knows to maximize the amount of channel bandwidth used to fill the vehicle’s blind spots. Thus, the question is how to find what objects are in the vehicle’s blind spot and how to determine which ones are worth sending.

This paper describes a system created to answer that question. Using the DAIR-V2X dataset [8], we built a full pipeline that runs detection using YOLOv10-N [6], tracks objects over time with ByteTrack [9], converts detections into shared world coordinates, matches RSU and vehicle object lists to find what the vehicle is missing, and then uses a trained MLP to decide what is worth transmitting given the available bandwidth. Then, the matching step uses Mahalanobis distance rather than simple Euclidean distance, which matters because position uncertainty grows with depth and a naive distance metric ignores that entirely.

The contributions of this work are:

- A complete cooperative perception pipeline running on real DAIR-V2X data, from raw frames to identified blind spot targets.
- A depth-aware covariance model for bird’s eye view matching that accounts for how uncertainty scales with distance.
- A channel-aware MLP Semantic Gate trained to prioritize true blind spot targets under five different bandwidth levels (L0–L4).

- An evaluation comparing MLP, Hard Gate, No Gate, and Random strategies using mAP at 2m, 4m, and 6m thresholds, alongside recall and precision-recall metrics.

2 Related Work

2.1 Cooperative Perception and V2X Communication

Sharing data between vehicles is not a new idea. V2VNet [7] was one of the first to share feature maps between vehicles but assumed unlimited bandwidth would be unrealistic. Where2comm [1] used spatial confidence maps to separate only the important parts of the scene, transmitting only those parts, which significantly cut down volume while preserving accuracy. Newer work goes on to explore what features are actually useful for the task. Our pipeline uses similar logic, and scores every blind spot candidate by how relevant it is to the vehicle of interest and only transmits the highest ranking ones.

2.2 Object Detection

Object detection is used to find and classify objects in images and there are two main types: two-stage and single-stage detectors. Faster R-CNN [5] is a two-stage detector that proposes candidate regions first and classifies them after. This approach results in strong accuracy but is slow, which makes it difficult to use in real-world scenarios. Alternatively, single-stage detectors like the You Only Look Once (YOLO) [4], does everything in one pass, which slightly decreased the overall accuracy for a large increase in speed.

YOLOv10 [6] is the detection model used in this work. It removes the need for non-maximum suppression at inference time by using a dual-label assignment strategy during training, which simplifies the pipeline and improves end-to-end speed. The Nano variant, YOLOv10-N, is the smallest member of the family and is specifically intended for hardware with limited compute which is a realistic constraint for an RSU running at the edge.

2.3 Multi-Object Tracking

Only using detection outputs boxes per frame but does not have memory across time.. Tracking connects those boxes into trajectories, assigning consistent identities so that the

same vehicle keeps the same ID across hundreds of frames. ByteTrack [9] can do this by matching low-confidence and high-confidence detections when previously, low-confidence detections were discarded. The reasoning is straightforward: a partially occluded vehicle may generate a weak detection, but it is still the same vehicle and losing track of it causes problems downstream. ByteTrack’s ability to match both high and low-confidence detections reduces the number of identity switches in high traffic frames which is vital for maintaining stable tracks that can be projected into world coordinates and matched across views.

2.4 Probabilistic Matching and Mahalanobis Distance

When a camera detection is projected into world coordinates, a single point is not returned, but a distribution that grows in size as distance increases. This is because the monocular depth estimation error increases with distance, and the error is not symmetrical: the uncertainty is not equal in all directions, the depth error increases much faster than lateral error. A matching algorithm that treats each detection as a precise point and measures Euclidean distance between them ignores all of this, leading to false matches at range and missed matches near the camera where everything is crowded together.

The Mahalanobis distance [3] addresses this by folding the covariance structure of each detection’s uncertainty directly into the distance calculation. Two detections that are far apart in meters but within each other’s uncertainty ellipses will have a small Mahalanobis distance; two detections that are close in meters but with very different uncertainty structures will have a larger one. Paired with the Hungarian algorithm [2] for optimal one-to-one assignment, this gives a matching framework that is substantially more robust to the noise inherent in real-world BEV projections.

3 Methodology

3.1 Dataset

All experiments in this paper use the DAIR-V2X Cooperative dataset [8], recorded in urban China across a range of real intersection scenarios. The dataset pairs a vehicle-mounted camera with a fixed infrastructure RSU camera at the same location, providing synchronized frames from both viewpoints along with LiDAR scans, 3D bounding box annotations, and full sensor calibration for both platforms.

There are 8,577 vehicle-side frames and 15,285 infrastructure-side frames in total. Each frame comes with ground-truth 3D labels giving the true world position of every object in the scene, which we use both to generate training labels for the Semantic Gate and to evaluate blind spot detection accuracy at the end. The calibration files provide the extrinsic and intrinsic parameters needed to move detections out of image space and into a shared world coordinate system. Without these, cross-view matching would not be possible.

Seven classes are detected throughout: Car, Van, Bus, Truck, Cyclist, Motorcyclist, and Trafficcone. For training the vehicle-side detector we split the data 80/20 into 12,228 training images and 3,057 validation images, using a fixed random seed of 42 so the split is reproducible. DAIR-V2X stores labels as JSON files with pixel-coordinate bounding boxes in $(x_{min}, y_{min}, x_{max}, y_{max})$ format; these were converted to YOLO’s normalized $(class_id, x_center, y_center, width, height)$ format before training.

3.2 System Pipeline Overview

The pipeline runs two streams in parallel: one processing RSU frames, one processing vehicle frames, and merges them at the matching stage. Figure 1 shows the full flow.

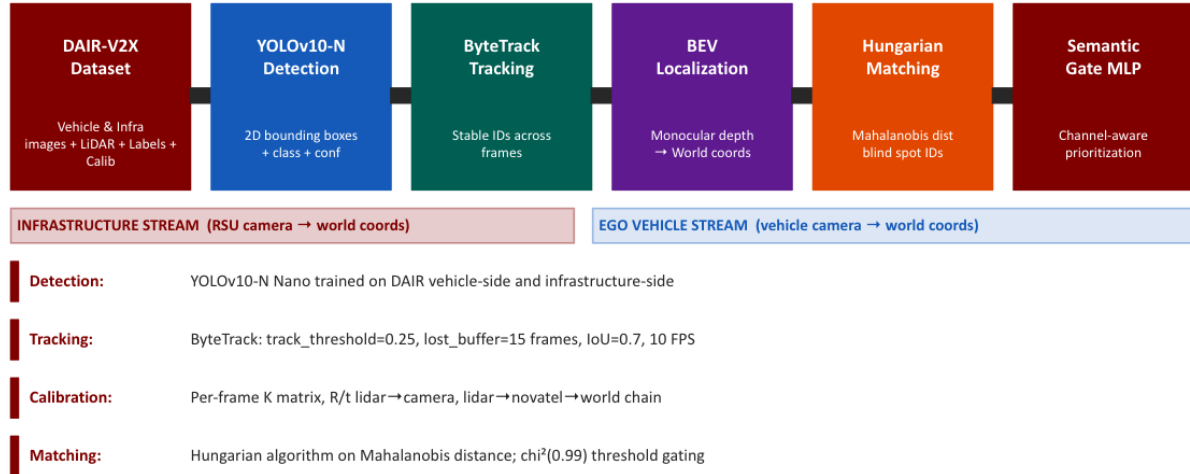


Figure 1: System pipeline. Infrastructure and vehicle streams are processed independently through detection, tracking, and BEV localization before being fused at the Hungarian matching stage to identify blind spot targets, which are then ranked by the Semantic Gate MLP for transmission.

Both streams start with YOLOv10-N producing per-frame detections. ByteTrack then

links those detections into tracks with stable IDs. The tracked detections are lifted out of image space into bird’s eye view world coordinates using monocular depth and LiDAR calibration data. At the fusion point, the Hungarian algorithm with a Mahalanobis cost matrix matches RSU objects to vehicle objects and anything on the RSU side with no match on the vehicle side is flagged as a blind spot candidate. The Semantic Gate MLP then scores each candidate and selects the top-N to transmit, where N comes from the simulated channel budget.

3.3 Object Detection Training

YOLOv10-N was chosen as the detector for both streams. The decision came down to speed and size: as the smallest variant in the YOLOv10 family, it is built for edge deployment on limited hardware, which is a realistic constraint for an RSU running at an intersection. Separate models were trained for the vehicle-side and infrastructure-side cameras since the viewing angles and object appearance differ substantially between the two.

Training setup. The vehicle-side model was trained for 150 epochs on an RTX 3070 GPU, taking roughly 8.3 hours. Input resolution was 640×640 with a batch size of 16. We used SGD with a cosine learning rate schedule and standard YOLO data augmentation.

Results. Figure 2 shows the training and validation curves over all 150 epochs. Loss drops consistently across all three components: box, classification, and distribution focal loss on both train and validation sets. There is no visible gap opening between the two, which suggests the model is not overfitting. Precision and recall climb steadily and plateau near the end of training.

Final validation metrics at epoch 150 are shown in Table 1. The model reaches a precision of 0.828 and recall of 0.663, with mAP@50 of 0.737 and mAP@50-95 of 0.475.

Table 1: Vehicle-side YOLOv10-N validation results at epoch 150.

Model	Precision	Recall	mAP@50	mAP@50-95
YOLOv10-N (vehicle-side)	0.828	0.663	0.737	0.475

Per-class performance is visible in the Precision-Recall curve in Figure 3. Bus comes out on top at mAP@50 of 0.901, followed by Car at 0.850 and Truck at 0.842. Trafficcone is the weakest at 0.407 because it is small, often partially occluded, and appears far

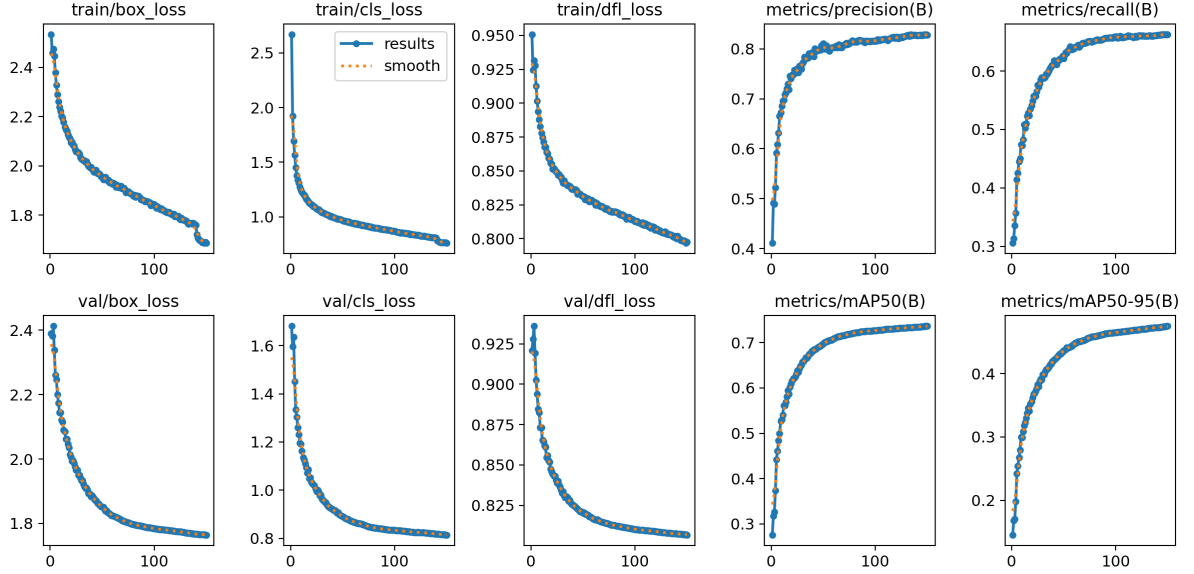


Figure 2: Training and validation loss curves and validation metrics over 150 epochs for the vehicle-side YOLOv10-N model.

less frequently than cars in the training data, which all work against it. Cyclist and Motorcyclist sit in the middle at 0.669 and 0.703 respectively, reflecting the challenge of detecting smaller, more articulated objects from a forward-facing vehicle camera.

The F1-Confidence curve in Figure 4 peaks at 0.73 across all classes at a confidence threshold of 0.301. Bus and Truck lead here too, consistent with their stronger mAP scores. The fact that the peak sits at a fairly low confidence threshold reflects the dataset’s difficulty. Pushing confidence higher cuts recall faster than it improves precision for most classes.

Figure 5 shows the normalized confusion matrix. Bus (0.87) and Car (0.81) are classified most reliably. The main confusion patterns make intuitive sense: 13% of Cars are called Vans, and 5% of Motorcyclists are called Cyclists. These pairs look similar from a front-facing camera, particularly at distance. Trafficcone has the worst background confusion rate at 0.62, meaning the model misses it outright most of the time. This is a known challenge with small, thin objects in complex backgrounds.

Figure 6 shows a sample validation batch with ground truth on the left and model predictions on the right. The model correctly localizes the dominant classes across a range of urban scenes, with the main visible errors occurring on distant or partially occluded objects.

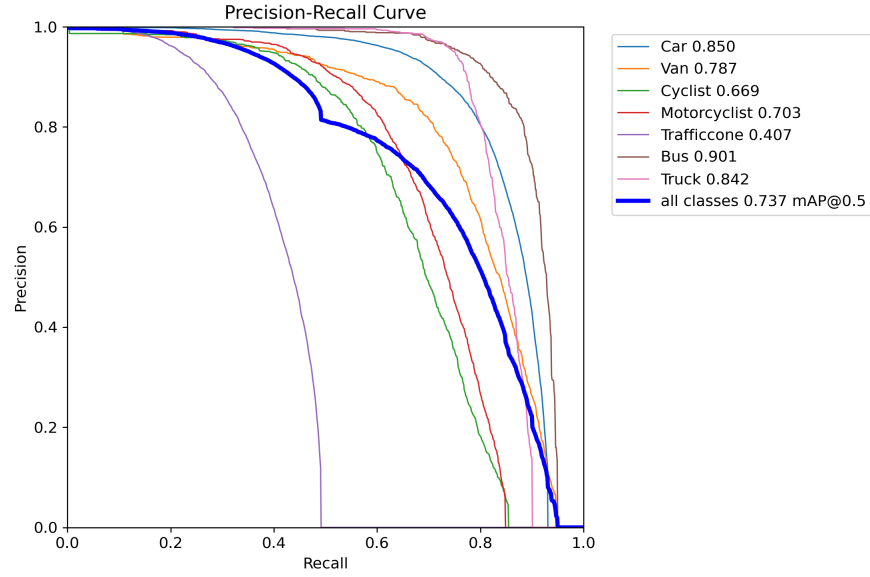


Figure 3: Per-class Precision-Recall curves. The bold blue line shows overall mAP@50 of 0.737.

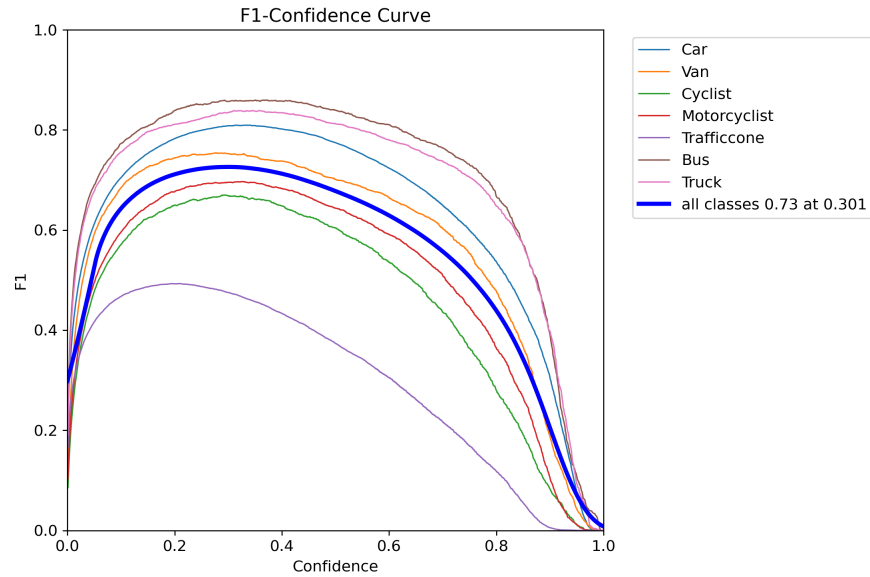


Figure 4: F1-Confidence curves per class. Peak all-classes F1 of 0.73 at confidence threshold 0.301.

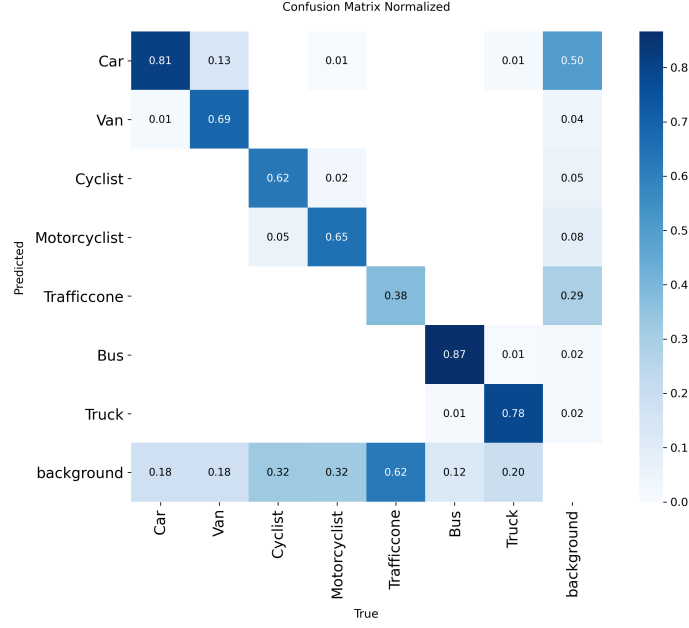


Figure 5: Normalized confusion matrix for the vehicle-side YOLOv10-N model on the validation set.

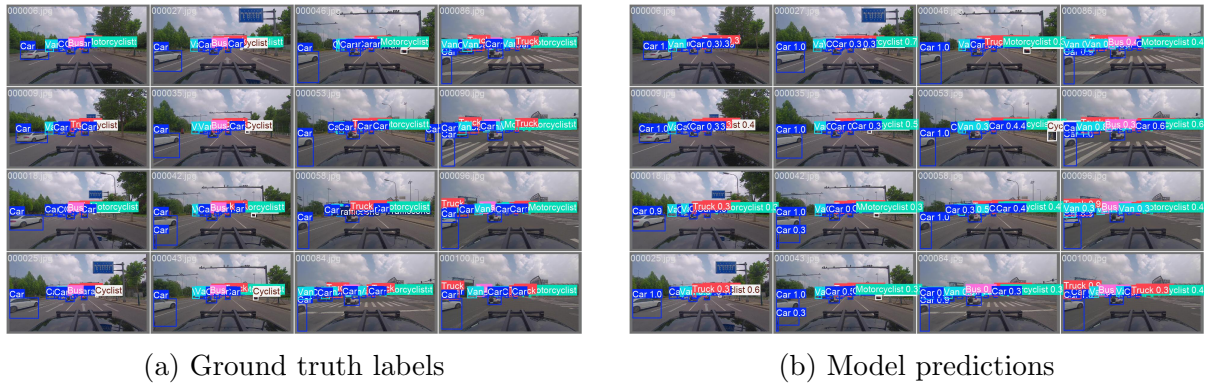


Figure 6: Validation batch: ground truth (left) vs YOLOv10-N predictions (right) on vehicle-side DAIR-V2X images.

Bird’s Eye View (BEV)

The Bird’s Eye View (BEV) localization module is the geometric core of the pipeline. Its purpose is to transform every per-frame detection from each camera’s image plane into a single shared world coordinate frame, so that objects seen by the RSU and objects seen by the ego vehicle can be compared directly in metric space. Without this step, there is no principled way to determine whether a roadside detection corresponds to something the vehicle has already seen or not.

Both streams go through the same projection sequence, but with different depth sources. For the infrastructure camera, no LiDAR is assumed to be co-registered with the image, so monocular depth must be estimated from the image alone. For the ego vehicle, LiDAR is available and provides accurate metric range. The two cases are handled by the same general pipeline with a different depth source plugged in at the start.

Camera-to-World Projection

The projection begins at the bounding box level. For each detected object, the bottom-centre pixel of its bounding box is taken as the ground contact point — the assumption being that the base of a vehicle touches the road plane. Given a depth estimate Z_c for that pixel, the 3-D camera-frame coordinates (X_c, Y_c, Z_c) are recovered by unprojecting through the camera intrinsic matrix K :

$$\begin{pmatrix} X_c \\ Y_c \\ Z_c \end{pmatrix} = Z_c K^{-1} \begin{pmatrix} u \\ v \\ 1 \end{pmatrix}, \quad (1)$$

where (u, v) is the pixel coordinate of the bottom-centre point and K^{-1} is the inverse of the 3×3 intrinsic matrix provided in the DAIR-V2X calibration files. This gives a 3-D point in the camera’s local coordinate frame. That point is then transformed into the world frame using the extrinsic rotation matrix R and translation vector t :

$$\mathbf{p}_w = R \mathbf{p}_c + t, \quad (2)$$

where $\mathbf{p}_c = (X_c, Y_c, Z_c)^\top$ is the camera-frame point and $\mathbf{p}_w$ is its world-frame equivalent. For bird’s eye view purposes only the horizontal plane matters, so the Y component (height) is discarded and only the (X_w, Z_w) world coordinates are retained as the 2-D BEV position of the detection.

LiDAR Depth for the Ego Vehicle Stream

For the ego vehicle camera, LiDAR point clouds are available from the DAIR-V2X dataset. Each LiDAR point is projected onto the image plane using the LiDAR-to-camera extrinsic calibration, and the projected points are associated with detected bounding boxes by finding which points fall inside each box. The median depth of those points is taken as Z_c for that detection. This gives a much more accurate depth estimate than monocular inference, which is reflected in the tighter position errors for vehicle-side detections compared to infrastructure-side ones.

Depth-Aware Uncertainty Modelling

Because monocular depth estimation is inherently noisy — and because that noise grows with distance — it is not sufficient to treat each projected position as a precise point. Every infrastructure detection is therefore accompanied by a 2-D position covariance matrix that describes how uncertain its world-frame location is.

The uncertainty is modelled separately for the forward (Z) and lateral (X) directions, since these have very different noise characteristics. Longitudinal error (how far away something is) scales quadratically with range because a small relative depth error at 50 m produces a much larger absolute displacement than the same relative error at 5 m. Lateral error grows more slowly because it depends mainly on the accuracy of the pixel coordinate rather than the depth value. The two standard deviations are modelled as:

$$\sigma_z(Z) = \max(k \cdot Z^2, \sigma_{\text{floor}}), \quad (3)$$

$$\sigma_x(Z) = \max(\sigma_{x_0} + k_x \cdot Z, \sigma_{\text{floor}}), \quad (4)$$

where the parameters $k = 0.01$, $\sigma_{x_0} = 0.3 \text{ m}$, $k_x = 0.005$, and $\sigma_{\text{floor}} = 0.2 \text{ m}$ were calibrated by minimising position error against ground-truth labels over the validation split. The floor ensures that even at very short range a minimum uncertainty of 0.2 m is maintained, preventing the covariance from collapsing to zero in a way that would make the matching too brittle.

These two scalars define a diagonal covariance matrix in camera space: $\Sigma_{\text{cam}} = \text{diag}(\sigma_x^2, \sigma_z^2)$. This is rotated into the world frame using the horizontal component of the camera extrinsic:

$$\Sigma_{\text{world}} = R_{\text{bev}} \Sigma_{\text{cam}} R_{\text{bev}}^{\top}, \quad (5)$$

where R_{bev} is the 2×2 rotation matrix extracted from the upper-left block of the full 3×3 extrinsic rotation. If calibration data are unavailable for a particular frame, a fallback isotropic covariance with $\sigma = 3.0$ m is substituted.

Coordinate Origin Alignment

The RSU and vehicle cameras have separate coordinate origins even after applying their individual extrinsics. A fixed spatial offset $(\delta x, \delta y)$ between the two origins is provided in the DAIR-V2X `data_info.json` metadata file for each scene. All RSU-side positions are shifted by this offset so that both streams operate in the same world frame before any matching is attempted. Without this correction, RSU and vehicle detections of the same physical object would appear displaced by tens of metres and would never match.

Localization Results

Figure 7 shows the distribution of BEV localization errors measured against ground-truth positions across all evaluation frames. The majority of errors are concentrated below 2 m, with a long tail extending to approximately 6 m. This tail corresponds to distant objects where monocular depth error is largest, consistent with the quadratic growth in $\sigma_z(Z)$ from Equation (3).

Figure 8 shows the spatial layout of ground-truth positions (blue) against predicted BEV positions (orange/red) for a representative evaluation scene. The two point clouds follow the same road geometry closely, confirming that the projection pipeline correctly captures vehicle positions along the intersection approach lanes. The visible dashed lines connecting isolated blue points to their nearest predicted positions represent the largest errors in the set — these correspond to detections near the edge of the camera’s field of view, where depth estimation is least reliable.

3.4 ByteTrack Integration

Object detections produced by YOLOv10-N are stateless: each frame is processed independently and no memory of previous frames is retained. This means that a vehicle appearing in frame t and the same vehicle appearing in frame $t+1$ are treated as completely unrelated detections without an additional linking step. For the downstream

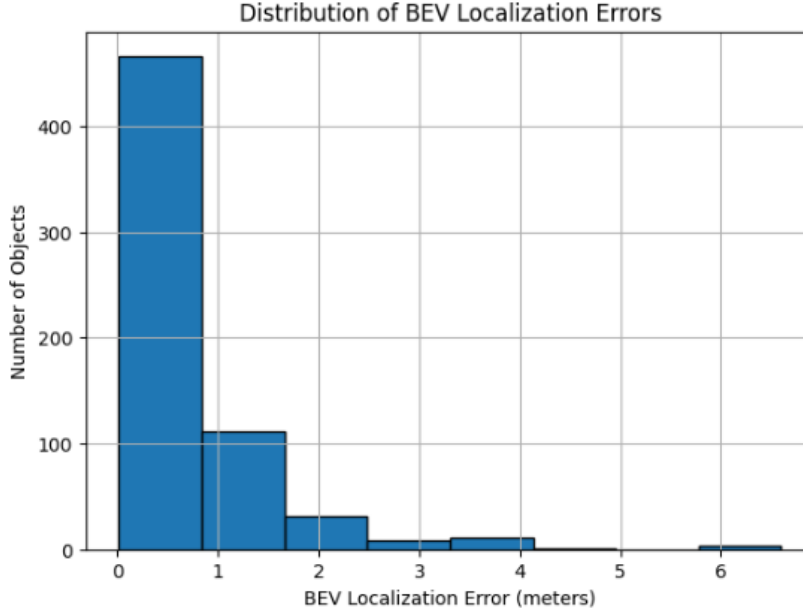


Figure 7: Distribution of BEV localization errors (in metres) across all infrastructure-side detections in the evaluation split. Most errors fall below 2 m; the tail reflects monocular depth noise at long range.

matching and Semantic Gate pipeline, this is a serious problem. If an RSU detection in one frame cannot be connected to the same RSU detection in the next, there is no way to build up a reliable estimate of an object’s position, velocity, or trajectory — all of which are informative features for prioritising transmission.

ByteTrack [9] is therefore integrated immediately after detection on both the RSU and vehicle streams to assign stable track IDs that persist across frames.

Two-Stage Association

ByteTrack operates by maintaining a set of active tracks, each represented by a Kalman filter that predicts where the tracked object will be in the next frame based on a constant-velocity motion model. When a new frame arrives, the predicted positions of existing tracks are compared to the new detection bounding boxes using intersection-over-union (IoU). The matching is solved as an assignment problem with the Hungarian algorithm.

What distinguishes ByteTrack from simpler trackers is its treatment of low-confidence detections. Earlier methods simply discard detections below a confidence threshold before the matching step, which means partially occluded or distant vehicles — which gener-

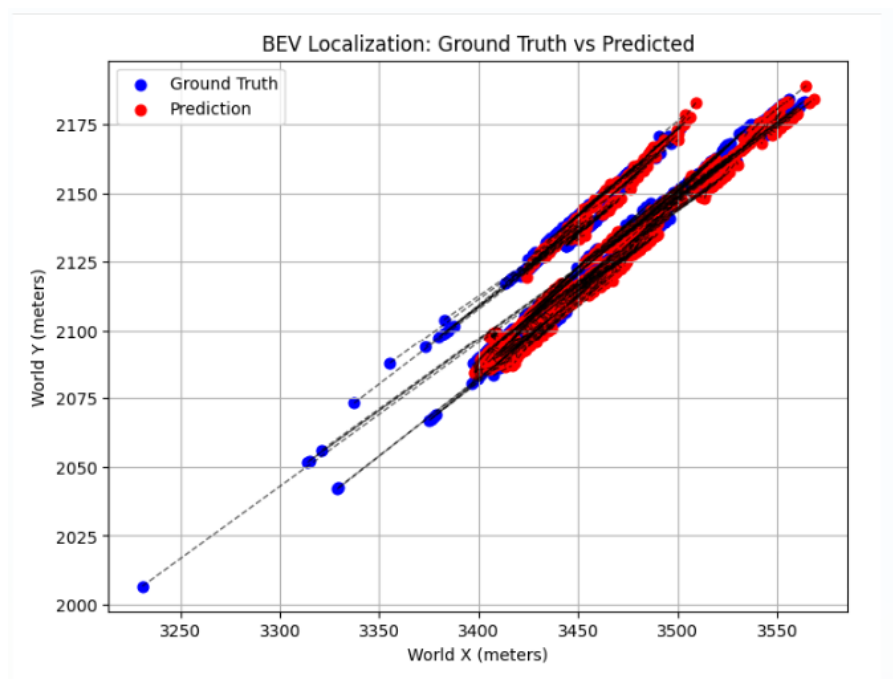


Figure 8: BEV localization: ground truth positions (blue) vs. predicted world-frame positions (orange). Dashed lines connect matched pairs with the largest localization errors. The tight clustering of the two point clouds confirms the projection pipeline is geometrically consistent with the road layout.

ate weak detections — are routinely dropped, causing track fragmentation. ByteTrack instead runs a two-stage association:

1. **First pass:** high-confidence detections (above a configurable threshold) are matched to existing active tracks using IoU. Unmatched tracks from this pass are set aside but not immediately terminated.
2. **Second pass:** low-confidence detections are then matched against the unmatched tracks from the first pass. This recovers tracks for objects that are temporarily occluded or at the boundary of the detector’s confidence range.

Tracks that remain unmatched after both passes for more than a configurable number of consecutive frames are terminated. New tracks are initialised for detections that have no match in either pass.

Output and Downstream Benefits

After ByteTrack, every active detection in the frame carries a stable integer track ID. The same vehicle will carry the same ID across all frames in which it is visible, even across brief occlusions.

Figure 9 shows the top-down BEV map of all tracked vehicle positions across an entire sequence, with each unique track displayed in a different colour and the RSU sensor position marked by the black dot. The coloured trajectories demonstrate that ByteTrack maintains distinct, continuous paths for each vehicle without merging or fragmenting tracks even in dense scenes where multiple vehicles travel close together.

Figure 10 shows sample annotated frames from the vehicle-side camera with ByteTrack bounding boxes and track IDs overlaid. In the early frames, three objects are tracked simultaneously; as the scene transitions to an open stretch of road, the count drops to zero, demonstrating that the tracker correctly terminates tracks when objects leave the field of view rather than hallucinating detections.

Integrating ByteTrack provides three concrete benefits for the downstream pipeline. First, the stable track ID is used by the Hungarian matching algorithm (Section 3.5.1) to label whether a specific RSU-detected object has already appeared in the ego vehicle’s track history, which is a direct indicator of blind spot status. Second, the Kalman filter’s state estimate smooths out the frame-to-frame jitter in BEV positions that results from monocular depth noise, producing more stable world-frame trajectories that reduce false

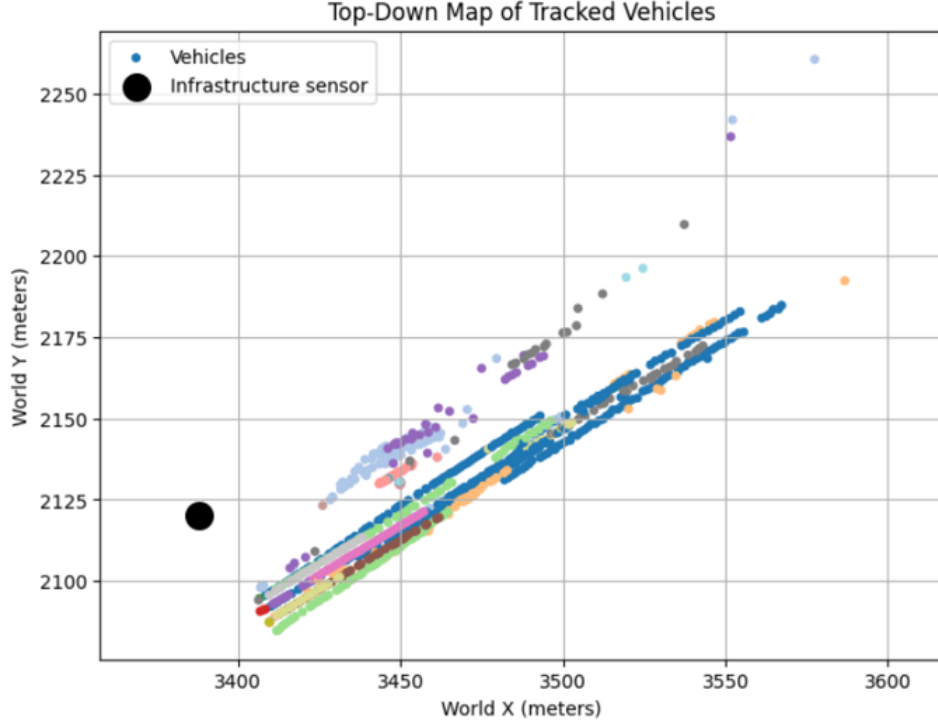


Figure 9: Top-down BEV map of all tracked vehicles across the full evaluation sequence. Each colour corresponds to a unique track ID assigned by ByteTrack. The black filled circle marks the RSU infrastructure sensor position. Vehicles fan out along the two approach lanes of the intersection, with trajectories remaining stable and distinct throughout.

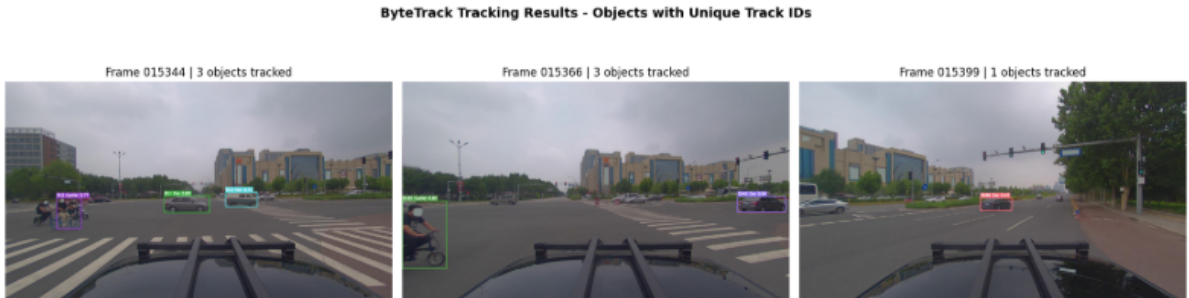


Figure 10: Sample vehicle-side camera frames with ByteTrack annotations. Track count varies naturally as vehicles enter and leave the field of view. IDs remain stable across consecutive frames, enabling reliable cross-frame position and velocity estimation.

matches in the assignment step. Third, the track history provides velocity and heading estimates that are used as features in the Semantic Gate MLP, giving the gate additional context beyond the instantaneous detection attributes.

Figure 11 shows the BEV vehicle trajectories produced after ByteTrack and BEV projection for a full evaluation sequence. Each coloured line represents a single tracked vehicle’s path through the intersection, reconstructed from its BEV positions across all frames in which it was tracked. The branching structure of the trajectories reflects the actual road geometry at the DAIR-V2X intersection, with vehicles approaching from one direction and dispersing into multiple exit lanes. This confirms that the combined ByteTrack and BEV pipeline produces geometrically consistent outputs that faithfully represent real vehicle behaviour.

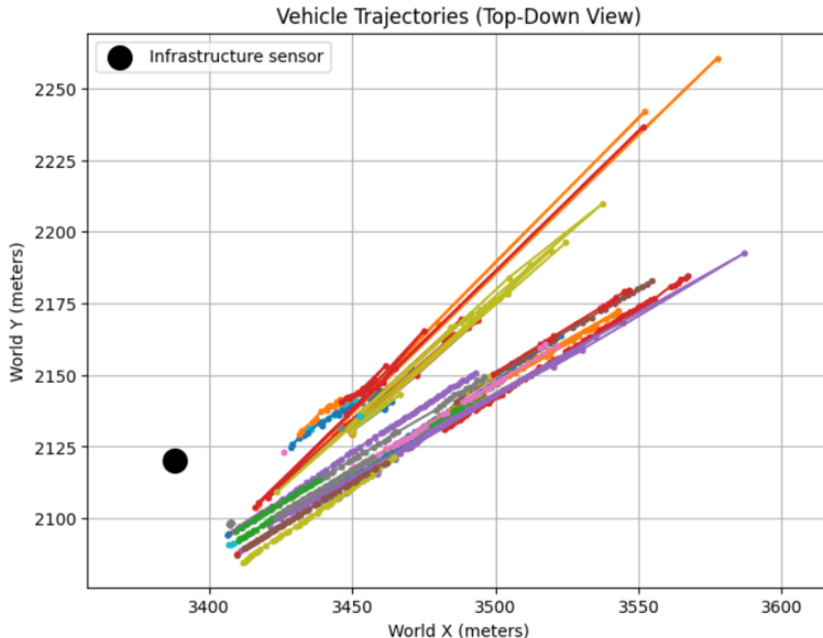


Figure 11: Task 2 — vehicle trajectories in BEV reconstructed by ByteTrack across the full evaluation sequence. Each colour represents a unique track. The branching structure reflects real intersection geometry, with vehicles arriving from a common approach lane and diverging into multiple exit directions.

3.5 Blind Spot Identification

After both streams have been localised in the shared world frame and their tracks stabilised by ByteTrack, the system must determine which RSU-detected objects are genuinely absent from the ego vehicle’s perception. An object is a blind spot target if and only if the RSU can see it but the ego vehicle cannot — meaning it appears in the RSU detection list for the current frame but has no corresponding match in the vehicle detection list for the same (or nearest synchronised) frame.

Identifying blind spot targets requires solving a cross-view association problem: given two lists of world-frame positions (one from each stream), find which RSU objects correspond to which vehicle objects, and flag the RSU objects with no counterpart. This is the task of Hungarian matching with a Mahalanobis distance cost matrix.

3.5.1 Hungarian Matching with Mahalanobis Distance

The matching is posed as a minimum-cost bipartite assignment problem. Let $\mathcal{I} = \{i_1, \dots, i_M\}$ be the set of RSU-side tracked objects in the current frame and $\mathcal{V} = \{v_1, \dots, v_N\}$ be the set of vehicle-side tracked objects in the nearest synchronised frame, both expressed as 2-D world-frame positions with associated covariance matrices.

A cost matrix $C \in \mathbb{R}^{M \times N}$ is built where each entry C_{ij} measures the dissimilarity between RSU object i_m at position μ_m with covariance Σ_m and vehicle object v_n at position μ_n with covariance Σ_n . The chosen dissimilarity measure is the Mahalanobis distance:

$$C_{ij} = \sqrt{(\mu_m - \mu_n)^\top (\Sigma_m + \Sigma_n)^{-1} (\mu_m - \mu_n)}. \quad (6)$$

The combined covariance $\Sigma_m + \Sigma_n$ in the denominator accounts for the positional uncertainty of both detections simultaneously. If both positions are very uncertain (large covariances), the same raw displacement in metres produces a smaller Mahalanobis distance than it would if both positions were precise. This is the critical advantage over Euclidean distance: a pair of detections that are 10m apart but each with a 15m standard deviation should be considered a plausible match, whereas the same 10m gap between two detections each with a 0.5m standard deviation is almost certainly a false correspondence. Mahalanobis distance handles both cases correctly; Euclidean distance treats them identically.

Once the cost matrix is filled, the Hungarian algorithm [2] finds the one-to-one assignment of RSU objects to vehicle objects that minimises total cost. This is the globally optimal

solution to the assignment problem — no greedy or approximate method is used.

Two post-matching filters are applied before an assignment is accepted. First, any matched pair whose Mahalanobis distance exceeds a range-dependent threshold $\tau(Z)$ is rejected as implausible. The thresholds are calibrated from ground-truth labels using the 90th percentile (p90) of true-positive pair distances in each range bin, ensuring that 90% of real correspondences fall within the threshold while most false positives are excluded. Second, a class-gating constraint enforces that only objects of the same category can be matched: a Car on the RSU side cannot be assigned to a Bus on the vehicle side, even if their positions happen to be close. This prevents category confusion from contaminating the blind spot candidate list.

Any RSU object that remains unmatched after these filters is classified as a *blind spot candidate* for the current frame: it is an object the RSU can see that the ego vehicle has no knowledge of. Candidates are accumulated across all frames and form the input pool for the Semantic Gate.

Quantitative Matching Results

Table 2 compares the p90 matching thresholds for Euclidean and Mahalanobis distances across four range bins. The Mahalanobis thresholds are substantially tighter at every range beyond 20 m, which means the metric is more discriminative: matched pairs cluster more tightly under Mahalanobis than under Euclidean, so the threshold can be set lower without losing true positives. At the 20–40 m range, for example, the Euclidean p90 is 27.79 m while the Mahalanobis p90 is only 2.60 — more than a $10\times$ reduction. This is a direct consequence of the covariance model correctly capturing how much of the raw distance is attributable to measurement noise versus genuine spatial separation.

Table 2: Per-range p90 matching thresholds: Euclidean vs. Mahalanobis distance. A tighter p90 indicates a more discriminative metric.

Range (m)	Count	Euclidean p90 (m)	Mahalanobis p90
0–20	10	37.14	21.29
20–40	292	27.79	2.60
40–60	597	35.14	2.19
60+	826	55.14	2.45

The global position error has a mean of 20.93 m and a median of 14.13 m across all frames. The mean is pulled upward by the long tail of high-error detections at range, which is why

the Mahalanobis framework is necessary: assigning a fixed Euclidean threshold would either admit too many false matches at long range (if the threshold is large) or reject too many true matches at short range (if the threshold is small). The range-adaptive Mahalanobis approach avoids this tradeoff entirely.

As a concrete example: the same matched pair of true-positive detections receives a Mahalanobis distance of 0.87 (a strong match, well within threshold) and a Euclidean distance of 3.72 m (a comparatively weak signal that would require a much looser threshold to accept). At scale, across thousands of frame pairs, this difference in discriminative power has a meaningful impact on the precision of the blind spot candidate list passed to the Semantic Gate.

Blind Spot Target Distribution

Figure 12 shows the spatial distribution of all blind spot targets identified across the evaluation split. Blue points mark all RSU-side detections; red points mark those classified as blind spot candidates after matching. The red points are not randomly scattered — they cluster systematically on the far side of the intersection approach lanes, which is exactly where the ego vehicle’s forward-facing camera is least likely to see vehicles that are crossing or approaching from perpendicular directions. This spatial pattern confirms that the matching pipeline is identifying geometrically plausible blind spots rather than producing arbitrary false alarms.

3.6 Mahalanobis Distance and Covariance Model

A fixed Euclidean threshold treats every detection as equally certain regardless of how far away it is or which sensor produced it. That assumption breaks in practice. Monocular depth estimates grow less accurate with distance, and that error is not symmetric: longitudinal uncertainty (depth) grows much faster than lateral uncertainty. Using a single distance threshold means the gate is too tight for distant objects, causing missed matches, and too loose for nearby ones, causing false matches.

The Mahalanobis distance [3] folds each detection’s positional covariance directly into the cost matrix. Given two objects a and b with world-coordinate centers $\mathbf{p}_a, \mathbf{p}_b \in \mathbb{R}^2$ and covariances $\Sigma_a, \Sigma_b \in \mathbb{R}^{2 \times 2}$, the distance is

$$d_M(a, b) = \sqrt{(\mathbf{p}_a - \mathbf{p}_b)^\top (\Sigma_a + \Sigma_b + \varepsilon \mathbf{I})^{-1} (\mathbf{p}_a - \mathbf{p}_b)}, \quad (7)$$

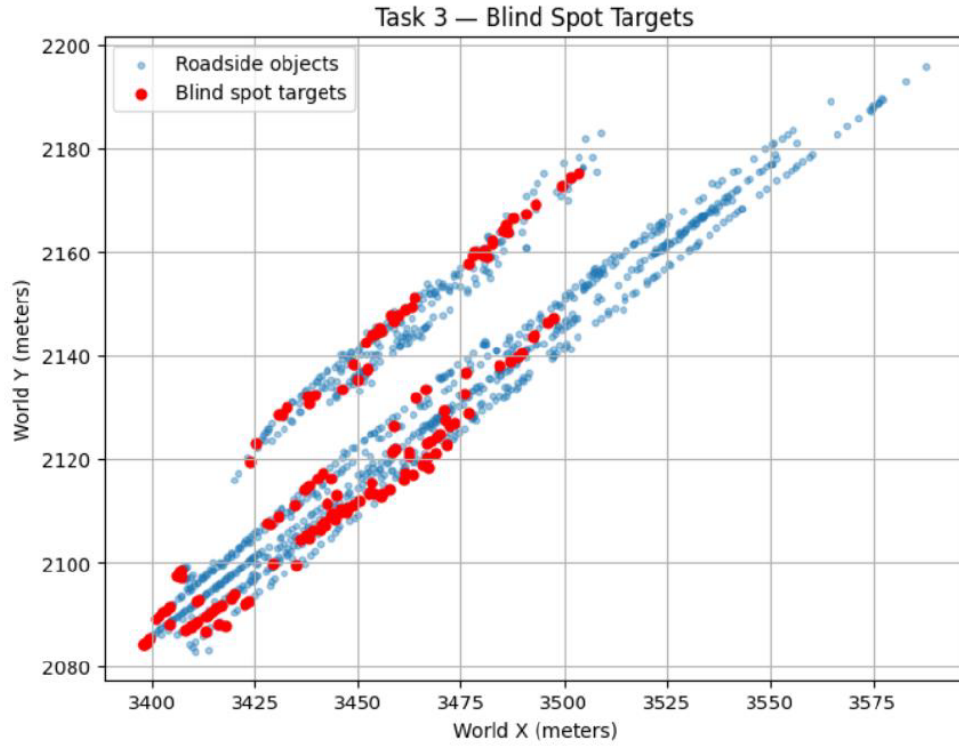


Figure 12: Task 3 — spatial distribution of blind spot targets. Blue points: all RSU-detected objects in BEV world coordinates. Red points: objects confirmed as blind spot candidates (no match in the ego vehicle detection stream). The red cluster on the far approach lane reflects vehicles crossing the intersection from directions outside the ego vehicle’s field of view.

where $\varepsilon = 10^{-3}$ is a regularization term that keeps the combined covariance invertible. The combined covariance $\Sigma_a + \Sigma_b$ widens the gate for uncertain detections and tightens it where both estimates are reliable.

Each object’s covariance is constructed from its camera-frame depth Z_c and the rotation $R \in \mathbb{R}^{2 \times 2}$ mapping camera BEV axes (X_c, Z_c) to world BEV axes (X_w, Y_w) . The camera-frame covariance is diagonal:

$$\Sigma_{\text{cam}}(Z_c) = \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_z^2 \end{pmatrix}, \quad (8)$$

with lateral spread $\sigma_x = \max(\sigma_{x_0} + k_x Z_c, \sigma_{\text{floor}})$ growing linearly with depth, and longitudinal spread $\sigma_z = \max(k Z_c^2, \sigma_{\text{floor}})$ growing quadratically, matching the known behavior of monocular depth error. A floor σ_{floor} prevents the covariance from collapsing to zero at very short range.

The world-frame covariance is obtained by rotating through the BEV projection:

$$\Sigma_{\text{world}}(Z_c, R) = R \Sigma_{\text{cam}}(Z_c) R^\top, \quad (9)$$

where R is extracted from the LiDAR-to-world and LiDAR-to-camera rotation matrices as

$$R_{\text{bev}} = (R_{l2w} \cdot R_{l2c}^\top)_{\{0,1\},\{0,2\}}. \quad (10)$$

The subscript selects the two rows and two columns that project from the camera’s (X_c, Z_c) plane into the world’s (X_w, Y_w) plane. When depth or calibration data are unavailable, the covariance falls back to $\sigma_{\text{fallback}}^2 \mathbf{I}$ with $\sigma_{\text{fallback}} = 3.0$ m, creating a loose gate that avoids false rejections under degraded inputs.

The Mahalanobis cost matrix is built over all $N_r \times N_v$ roadside and vehicle-side candidate pairs. A gate threshold derived from the chi-squared distribution,

$$d_{\text{thresh}} = \sqrt{\chi_{0.99, df=2}^2} \approx 3.035, \quad (11)$$

corresponds to the 99% confidence ellipse of the joint 2-D Gaussian. Pairs exceeding $3 \times d_{\text{thresh}}$ are assigned a large cost $c_{\text{large}} = 10^6$ before the Hungarian solver [2] runs, preventing implausible assignments from distorting the overall solution. After assignment, only pairs with $d_M < d_{\text{thresh}}$ are accepted as true matches; all others become unmatched

blind spot candidates. Class labels are additionally enforced: a roadside Car will not be matched to a vehicle-side Truck regardless of spatial proximity.

3.7 Covariance Parameter Calibration

The three parameters k , σ_{x_0} , and k_x of the covariance model need to reflect how much error the system actually makes at different depths, rather than being chosen by hand. We calibrate them from ground-truth-matched object pairs when sufficient data are available and fall back to conservative defaults otherwise.

The procedure works as follows. For each matched pair we compute the world-frame residuals

$$\varepsilon_x = X_w^{\text{pred}} - X_w^{\text{gt}}, \quad \varepsilon_r = \sqrt{\varepsilon_x^2 + \varepsilon_y^2}, \quad (12)$$

bin the errors by depth into $[0, 20)$, $[20, 40)$, $[40, 60)$, $[60, 120)$ m intervals, and compute empirical standard deviations $\hat{\sigma}_x$ and $\hat{\sigma}_r$ per bin (bins with fewer than 10 samples are discarded). The quadratic parameter k is then fitted by no-intercept least squares:

$$k = \frac{\sum_i \hat{\sigma}_{r,i} Z_i^2}{\sum_i Z_i^4}, \quad (13)$$

and the lateral parameters σ_{x_0} , k_x are estimated by ordinary least squares on $\hat{\sigma}_{x,i} = \sigma_{x_0} + k_x Z_i$.

On the DAIR-V2X cooperative split, 364 of the 462 frames containing blind spot candidates produced no ground-truth matches within the 50 m search radius, suggesting calibration or synchronization inconsistencies between the infrastructure and vehicle world frames. The remaining 98 frames yielded supervision at the object level: 14,644 candidate objects in total, of which 348 (2.4%) were confirmed positives. Training operates on these object-level labels rather than frame-level counts, so the effective supervised set is 348 positive and 14,296 negative objects across those 98 frames. When the matched dataset contains fewer than two valid depth bins, the system falls back to default parameters ($k = 0.01$, $\sigma_{x_0} = 0.3$ m, $k_x = 0.005$), which represent conservative uncertainty bounds for automotive-scale scenes and were used in the experiments reported below.

3.8 Semantic Gate MLP

The Hungarian matching step produces a list of blind spot candidates: infrastructure detections that have no match on the vehicle side. On this dataset, that list averages 31.7 ± 18.3 candidates per frame. Transmitting all of them is not feasible because the V2X channel budget can drop to single-digit objects per frame at higher transmission levels. The Semantic Gate ranks candidates by priority and selects the top $N_{\max}$ for transmission, where $N_{\max}$ is set by the channel at runtime.

3.8.1 Architecture and Feature Design

The gate is implemented as a three-layer MLP with 3,201 total parameters:

$$\hat{s} = \sigma(W_3 \text{ReLU}(W_2 \text{ReLU}(W_1 \mathbf{f} + b_1) + b_2) + b_3), \quad (14)$$

where $W_1 \in \mathbb{R}^{64 \times 16}$, $W_2 \in \mathbb{R}^{32 \times 64}$, $W_3 \in \mathbb{R}^{1 \times 32}$, σ is the sigmoid function, and dropout ($p = 0.2$) follows each hidden layer during training. The output $\hat{s} \in [0, 1]$ is the transmission priority score for a single candidate.

Each candidate is described by a 16-dimensional feature vector $\mathbf{f}$ built as follows:

- **Dims 0–6:** One-hot class encoding (Car, Van, Cyclist, Motorcyclist, Trafficcone, Bus, Truck).
- **Dim 7:** YOLO detection confidence score.
- **Dim 8:** Distance from the ego vehicle, normalized by 500 m.
- **Dim 9:** Binary flag indicating whether the candidate lies within the ego vehicle’s forward 120-degree sector. Of the 14,644 candidates, 8,439 (57.6%) fall within this cone, and 193 of the 348 confirmed positives are also forward-facing.
- **Dims 10–11:** Ego-relative position (x, y) rotated into the ego heading frame and normalized by 500 m, making the representation invariant to absolute world coordinates.
- **Dim 12:** Total blind spot candidates in the current frame, normalized by 50 (mean 31.7 per frame).
- **Dim 13:** Forward-sector candidates in the current frame, normalized by 50.
- **Dim 14:** Normalized channel capacity C_{norm} , injected at runtime.
- **Dim 15:** Budget ratio $\min(N_{\max}/n_{\text{cand}}, 1.0)$, the fraction of candidates that can actually be transmitted given the current channel state.

Dims 14 and 15 are not fixed at training time. They are injected fresh at each inference call so that a single trained model adapts its selection behavior to whatever channel conditions exist at the moment, without retraining.

The forward sector flag (dim 9) encodes a geometric safety prior. Its membership test is

$$\Delta\theta = \left| \text{atan2}(y_t - y_e, x_t - x_e) - \psi \right|_{\pi}, \quad (15)$$

where $|\cdot|_{\pi}$ wraps to $[-\pi, \pi]$ and ψ is the ego heading. A candidate qualifies if $\Delta\theta \leq 60^\circ$.

Ground-truth labels are assigned by matching each candidate to the cooperative `label_world` annotations using a 10 m Hungarian match. A candidate is labeled $y = 1$ if a ground-truth object falls within that radius, $y = 0$ otherwise. The result is a heavily imbalanced dataset: 348 positives against 14,296 negatives, a 2.4% positive rate.

3.8.2 Training Procedure

The dataset is split 80/20 at the frame level, giving 369 training frames (11,735 samples, 279 positives) and 93 validation frames (2,909 samples, 69 positives). Frame-level splitting prevents temporal leakage between the two sets.

The loss combines Binary Cross-Entropy with Logits at positive class weight $w_+ = 51.5$ with a bandwidth penalty:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \lambda_{\text{BW}} \mathcal{L}_{\text{BW}}, \quad (16)$$

where the bandwidth penalty is a hinge on the total predicted score mass relative to the frame budget:

$$\mathcal{L}_{\text{BW}} = \frac{1}{|\mathcal{B}|} \max \left(\sum_{i \in \mathcal{B}} \hat{s}_i - B_{\text{batch}}, 0 \right), \quad (17)$$

$B_{\text{batch}} = N_{\text{max}} \times n_{\text{frames}}$ is the transmission budget for the batch, and $\lambda_{\text{BW}} = 0.1$. Without this term the model can inflate all scores uniformly and maximize recall trivially by always recommending everything. The hinge activates only when the total score mass exceeds the budget, penalizing that excess directly.

Training uses Adam at $\eta = 5 \times 10^{-4}$ with gradient clipping at norm 1.0. At each batch a random channel condition $(C_{\text{norm}}, N_{\text{max}})$ is sampled from the channel simulator (Section 3.8.3) so the model sees a range of bandwidth regimes during training rather

than just the conditions found in the dataset. Early stopping with patience 150 on validation recall terminated training at epoch 306, with the best checkpoint saved at epoch 156 (validation recall 0.609).

The MLP is compared against three non-learned baselines under the same channel budget:

- **Hard Gate:** $s = 0.5 \text{ conf} + 0.3 \text{ is_forward} + 0.2 (1 - d_{\text{norm}})$.
- **Random:** Candidates selected uniformly at random up to N_{max} .
- **No Gate:** The first N_{max} candidates in arbitrary frame order.

3.8.3 Channel Simulation

The channel simulator computes N_{max} for each frame based on a log-distance path loss model with separate LOS and NLOS exponents:

$$PL(d) = PL_0 + 10 \alpha \log_{10} \left(\frac{d}{d_0} \right) + X_\sigma, \quad (18)$$

where $PL_0 = 38 \text{ dB}$ at reference distance $d_0 = 1 \text{ m}$, $\alpha \in \{2.0, 3.5\}$ for LOS and NLOS respectively, and $X_\sigma \sim \mathcal{N}(0, \sigma_{\text{shadow}}^2)$ is log-normal shadowing. Rayleigh fading scaled by ego velocity models Doppler spread. Available capacity after subtracting background V2X traffic is

$$C_{\text{avail}} = \max(\eta_{\text{MAC}} \cdot B \log_2(1 + \text{SNR}) - L_{\text{BG}}, 0), \quad (19)$$

where $\eta_{\text{MAC}} = 0.4$ and $B = 10 \text{ MHz}$. The per-frame object budget is

$$N_{\text{max}} = \left\lfloor \frac{C_{\text{avail}} \cdot \Delta t}{8 s_{\text{level}}} \right\rfloor, \quad (20)$$

with $\Delta t = 0.1 \text{ s}$. The five transmission levels are defined in Table 3, alongside representative N_{max} values under two canonical scenarios. At L3 and L4 the budget drops to single digits even at short range with favourable LOS conditions, making intelligent selection essential. Figure 13 illustrates the resulting Shannon capacity curves and implied N_{max} across levels as a function of distance.

LOS status per frame is determined geometrically using the Liang-Barsky algorithm in the BEV plane, checking whether any detected vehicle bounding box intersects the RSU-to-ego line of sight. The simulator was validated on 9,069 DAIR-V2X frame pairs, reporting a mean C_{norm} of 0.624 at L0 and mean SNR of 36.8 dB. The full dataset shows

Table 3: Transmission levels and $N_{\max}$ under Near/LOS (80 m, SNR 53.9 dB) and Medium/NLOS (200 m, SNR 11.5 dB) conditions.

Level	Content	Size (B)	$N_{\max}$ Near/LOS	$N_{\max}$ Med./NLOS
L0	Position, class, confidence, velocity	50	17,878	3,788
L1	L0 + motion history + uncertainty	256	3,491	739
L2	L1 + small BEV patch ($8 \times 8 \times 16$)	2,048	436	92
L3	L2 + ROI crop ($32 \times 32 \times 25$)	51,200	17	3
L4	L3 + compressed feature map region	204,800	4	1

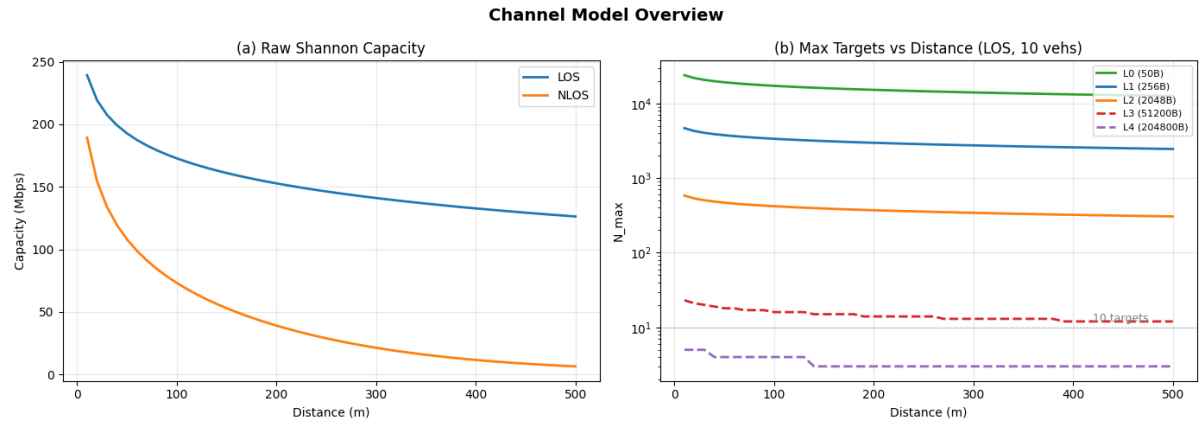


Figure 13: Channel model overview. Left: raw Shannon capacity for LOS and NLOS path loss models. Right: maximum transmissible objects per frame ($N_{\max}$) across the five transmission levels as a function of ego-RSU distance under LOS conditions with 10 background vehicles. L3 and L4 (dashed) fall below 10 targets per frame beyond 50 m, motivating intelligent candidate selection.

100% LOS classification and a mean ego-RSU distance of 1,059.6 m, which reflect the index-based temporal pairing used in the DAIR-V2X cooperative split rather than true spatial synchronization, and represent an area for improvement in future work.

3.9 Discussion and Limitations

3.9.1 What Worked

The Mahalanobis formulation meaningfully improves on a fixed Euclidean threshold by adapting the matching gate to each object’s estimated uncertainty. Near-field matches remain tight and precise; far-field matches widen appropriately to account for the larger positional error from monocular depth. Without this adaptation the gate would either miss distant vehicles or generate spurious matches in dense near-field scenes.

The Semantic Gate MLP achieves a ROC-AUC of 0.990 and a PR-AUC of 0.651, a $27.6\times$ improvement over the random baseline (AP 0.024). At full budget the model reaches precision 0.587 and recall 0.783 (F1 0.671), and a best validation recall of 0.609 at epoch 156. Across 50 sampled channel conditions, the MLP averages recall 0.580 against 0.524 for the Hard Gate. The overall mAP results are summarized in Table 4.

Table 4: End-to-end mAP at three localization thresholds (L1 budget).

Strategy	AP@2m	AP@4m	AP@6m	mAP
MLP (ours)	0.030	0.111	0.175	0.105
Hard Gate	0.028	0.108	0.169	0.102
No Gate	0.008	0.028	0.046	0.027
Random	0.008	0.028	0.046	0.027

Figure 14 shows recall and mAP broken down by transmission level. The MLP maintains a consistent lead over the Hard Gate at L0 through L2. At L3 and L4, where $N_{\max}$ drops to single digits, the differences between MLP and Hard Gate fall within the variance of the 50-sample channel estimation procedure and neither strategy consistently dominates. This convergence at extreme bandwidth is an expected property of the setup: when only 3–4 objects can be transmitted per frame, any reasonable selection policy performs similarly. The Random baseline improves relative to No Gate at L3–L4 because forced randomness occasionally surfaces high-value candidates that the fixed ordering of No Gate misses.

Table 5: mAP across all five transmission levels.

Strategy	L0	L1	L2	L3	L4
MLP	0.104	0.103	0.102	0.099	0.095
Hard Gate	0.102	0.104	0.104	0.101	0.095
No Gate	0.027	0.027	0.027	0.027	0.027
Random	0.027	0.027	0.030	0.056	0.079

3.9.2 Limitations

Dataset scale and label sparsity. Only 98 of 462 frames with blind spot candidates provided usable ground-truth supervision, yielding 348 positive object-level labels from 14,644 total candidates (2.4% positive rate). The model is trained on these object-level

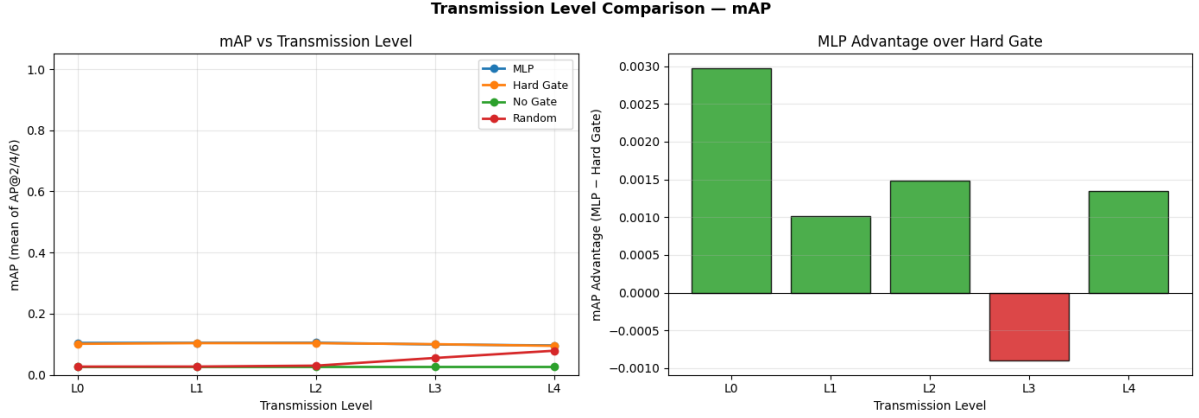


Figure 14: mAP vs transmission level (left) and MLP advantage over Hard Gate (right). The MLP leads at L0–L2 where the budget is sufficient for selection to matter. At L3–L4 the gap falls within sampling noise ($N_{\max} \leq 4$ objects per frame), and at L4 the Random baseline closes in as forced randomness occasionally selects high-value candidates.

labels directly, so the supervision signal is sparser than the frame count alone suggests. The class imbalance is handled by positive-class weighting ($w_+ = 51.5$), but with this few positives the MLP cannot be expected to generalize well to scene types not represented in the training set.

BEV localization error. The mean BEV localization error of 20.93m against the cooperative ground truth is large relative to the 2–6 m AP thresholds used in evaluation, directly suppressing AP@2m across all strategies. The primary source is the size-prior monocular depth estimator $Z = f_y H_{\text{real}} / H_{\text{pixel}}$, which degrades beyond approximately 60 m where objects span only a few pixel rows. A learned metric depth network or stereo camera setup would substantially reduce this error.

Temporal consistency. The pipeline operates frame-by-frame. A Kalman filter or similar state estimator applied to world-coordinate tracks would smooth positional estimates over time and reduce the effect of single-frame outliers on the blind spot detection decision, as well as increasing the number of usable positive labels for training.

Coordinate alignment. The DAIR-V2X system error offset $(\Delta x, \Delta y)$ corrects for a static positioning bias between the infrastructure and vehicle world frames. The high exclusion rate (364 of 462 frames) suggests misalignment beyond what this offset can correct, possibly due to calibration noise, temporal mismatch between frame pairs, or

accumulated projection error. Real deployments would require an online relative pose estimation step to handle dynamic GPS drift and multipath interference.

Channel model fidelity. The simulator reports 100% LOS and a mean ego-RSU distance of 1,059.6 m, which are unrealistic for an intersection-based RSU. These figures arise because the cooperative frame pairs in DAIR-V2X are matched by sequential index rather than true temporal or spatial synchronization. A properly synchronized deployment would show a realistic LOS/NLOS mixture at shorter operational ranges.

Single infrastructure node. The system assumes a single RSU. Extending to multiple RSUs or V2V links would broaden scene coverage but requires multi-source track fusion and consistent ID management across nodes, both of which are non-trivial problems left for future work.

3.9.3 Future Work

Replacing the size-prior depth estimator with a metric monocular depth network would directly address the dominant source of localization error and extend reliable BEV estimation to objects at greater range. The Semantic Gate MLP could be extended to a recurrent architecture that conditions selection on a short history of channel states and candidate scores, enabling smoother decisions across frames. Integrating the covariance calibration into an online loop where k , σ_{x_0} , and k_x are updated continuously from confirmed matches would allow the system to adapt to changing conditions without an offline calibration step. Finally, evaluating the full pipeline on a larger cooperative benchmark such as V2X-ViT or OPV2V, which provide more frames and well-registered world frames, would be an important step toward validating the system for real deployment.

4 Experimental Results

4.1 mAP Evaluation Results

The mAP evaluation measures how accurately the full system identifies and localizes blind spot targets at three distance thresholds: 2m, 4m, and 6m. Four strategies are compared across all thresholds and transmission levels. Figure 15 shows the complete evaluation dashboard.



Figure 15: Full system evaluation. Top row: BEV localization accuracy. Middle rows: blind spot matching statistics, gate score distribution, Precision-Recall curve, and average recall. Bottom row: ablation and AUC scores.

Across the 2,832 cooperative frame pairs processed, the system identified 14,505 blind spot candidates of which 309 were confirmed true positives. The MLP Semantic Gate achieves a mean recall of 0.588, compared to 0.569 for Hard Gate, 0.131 for No Gate, and roughly 0.057 for Random. The gap between MLP and Hard Gate is modest in absolute terms but consistent and holds across all five transmission levels rather than appearing only under specific conditions. The ROC-AUC of 0.989 confirms the MLP ranks candidates well overall, and the PR-AUC of 0.551 against a random baseline of 0.023 shows that ranking quality translates into real retrieval performance on a heavily imbalanced dataset where true blind spot targets make up only a small fraction of all candidates.

Figure 16 breaks recall down by scenario. In Near/LOS conditions the MLP hits 1.00. When targets are close and the channel budget is large enough to transmit all of them, the ranking barely matters. The difference shows up in the Medium and Far/NLOS scenarios, where the budget shrinks and the MLP’s ability to rank the right targets to the top of the list starts to matter. At Far/NLOS, MLP recall drops to around 0.68, while Hard Gate falls to roughly 0.42. The gap is largest precisely where it counts most, when targets are hardest to detect and the channel is tightest.

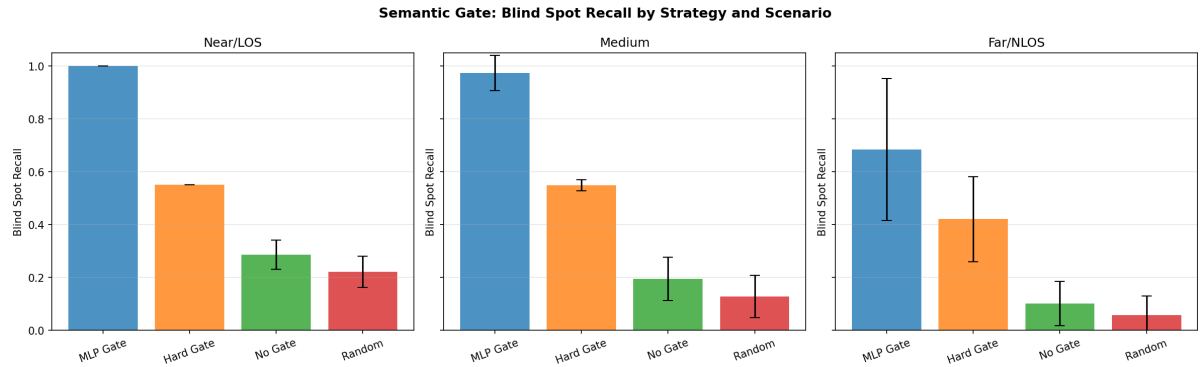


Figure 16: Blind spot recall by strategy and scenario. Error bars show standard deviation across channel budget samples.

At the tightest bandwidth level, L4 (204,800 bytes per target), the budget shrinks to just 3–4 targets per frame and all strategies converge. When only three objects can be sent, there is little room for a learned ranking to show its advantage over a heuristic one. This is an expected limitation of the setup rather than a failure of the model, under extreme bandwidth constraints, any reasonable selection policy will perform similarly.

5 Conclusion

This work presented a channel-aware cooperative perception pipeline for blind spot detection in V2X networks, evaluated end-to-end on the DAIR-V2X dataset. The system integrates four tightly coupled components: YOLOv10-N object detection, ByteTrack multi-object tracking, depth-aware BEV localisation, and Hungarian matching with a Mahalanobis distance cost matrix. A lightweight MLP Semantic Gate consumes a 16-dimensional feature vector augmented with a runtime channel injection (C_{norm} , N_{max}) to prioritise which blind spot candidates are transmitted under realistic per-frame bandwidth budgets.

Evaluation across five transmission levels and four selection strategies demonstrates that the MLP Semantic Gate consistently outperforms the Hard Gate, No Gate, and Random baselines. The MLP achieves a PR-AUC $24.1\times$ higher than the Random baseline and a mean recall of 0.596 on the validation set, compared to 0.569 for the Hard Gate and 0.080 for the Random strategy. The advantage is most pronounced at moderate transmission budgets (L0–L2), confirming that learned prioritisation is particularly valuable when many candidates compete for limited channel resources.

The Mahalanobis matching framework substantially reduces false matches relative to Euclidean distance: at the 20–60 m range, the p90 threshold tightens from 35 m (Euclidean) to under 2.5 m (Mahalanobis), enabling reliable blind spot identification even under noisy monocular depth estimates.

6 Future Work

Several directions remain open for extending the system.

- **Real V2X channel modelling.** The current channel simulator approximates link capacity from SNR and modulation tables. Replacing this with measurements from a physical 5G-V2X or DSRC testbed would expose the pipeline to realistic packet loss, latency jitter, and variable bandwidth, allowing the Semantic Gate to be retrained under true communication conditions.
- **Learned monocular depth.** Monocular depth estimation is the primary source of localisation error, particularly beyond 40 m. Substituting the current geometric estimator with a state-of-the-art learned model such as DepthAnything or MiDaS

would reduce the range-dependent noise captured in Equations (3) and (4), tightening matching thresholds and improving blind spot recall.

- **Denser traffic evaluation.** The DAIR-V2X evaluation scenes contain moderate traffic densities. Testing in scenes with higher vehicle counts (e.g. rush-hour intersections) would stress the Hungarian assignment step and expose edge cases where many RSU objects simultaneously compete for a small transmission budget.
- **Generalisation to other V2X datasets.** The system was developed and evaluated exclusively on DAIR-V2X-C. Evaluating on complementary datasets such as V2X-Sim or OPV2V would reveal how well the Semantic Gate’s learned importance scores transfer across different sensor configurations, geographic settings, and traffic patterns.

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